6.18 DECISION MATHEMATICS 2, D2 (4772) A2

Objectives

To give students experience of modelling and of the use of algorithms in a variety of situations.

To develop modelling skills.

The problems presented are diverse and require flexibility of approach. Students are expected to consider the success of their modelling, and to appreciate the limitations of their solutions.

Assessment

Examination (72 marks)
1 hour 30 minutes
Candidates answer all four questions.

Two questions, each worth about 16 marks and
two questions, each worth about 20 marks.

Assumed Knowledge

Candidates are expected to know the content of *C1* and *C2* and *D1*.

Calculators

In the MEI Structured Mathematics specification, no calculator is allowed in the examination for *C1*. For all other units, including this one, a graphical calculator is allowed.

DECISION MATHEMATICS 2, D2		
Specification	**Ref.**	**Competence Statements**

LINEAR PROGRAMMING		
The simplex algorithm.	D2L1	Be able to solve simple maximisation problems with ≤ constraints and with two or more variables.
Geometric interpretation.	2	Be able to identify tableaux (initial, intermediate and final) with feasible points, particularly in the case of problems involving two or three variables.
≥ inequalities.	3	Know and be able to apply the two-stage simplex and the big-M methods to construct an initial feasible solution to problems involving ≥ constraints.
Equality constraints.	4	Understand how to model an equality constraint by using a pair of inequality constraints.
Problem solving.	5	Be able to formulate and solve a variety of problems as linear programming problems, and be able to interpret the solutions.

NETWORKS		
The shortest path between any two nodes in a connected Network.	D2N1	Know and be able to apply Dijkstra's algorithm repeatedly.
	2	Know and be able to apply Floyd's algorithm.
	3	Be able to analyse the complexity of Dijkstra’s and Floyd’s algorithms.
The travelling salesperson problem (TSP).	4	Be able to convert the practical problem into the classical problem.
	5	Be able to interpret a solution to the classical problem in terms of a solution to an underlying practical problem.
	6	Be able to analyse the complexity of complete enumeration.
	7	Be able to construct an upper bound for the solution to the classical problem.
	8	Be able to construct a lower bound for the solution to the classical problem.
The route inspection (Chinese postperson) problem (CPP).	9	Know and be able to apply the route inspection algorithm.
	10	Be able to analyse the complexity of the algorithm.

DECISION MATHEMATICS 2, D2			
Ref.	**Notes**	**Notation**	**Exclusions**

LINEAR PROGRAMMING			
D2L1	The tabular form of algorithm.		Any consideration of the complexity of the algorithm.
2	Showing alternative feasible points and their costs.		
3	Including the possibility that there is no feasible solution.		
4			
5	Problems will not be restricted to maximisation problems with ≤ constraints, such as production planning problems involving the maximisation of profit or contribution subject to resource constraints. Instead they will encompass blending problems, shortest path problems, stock cutting problems, etc.		Non-linear problems, methods for solving integer programming problems.

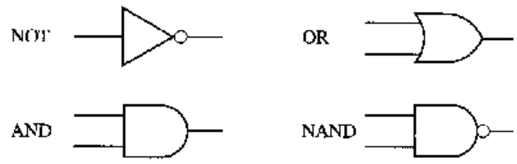
NETWORKS			
D2N1			
2			
3	Floyd and the repeated application of Dijkstra both have cubic complexity.		
4	Practical problem: revisiting vertices allowed; network not necessarily complete.		
5	Classical problem: no revisiting allowed; network complete.		
6	Problems of factorial complexity.		
7	Using the nearest neighbour algorithm.		Tour-to-tour improvement algorithms.
8	Using a minimum connector of a reduced network.		
9	Pairing of odd nodes and repeating shortest paths between the members of each pair.		
10	$\frac{(2n)!}{2^n n!}$ ways of pairing $2n$ odd nodes.		

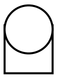
DECISION MATHEMATICS 2, D2		
Specification	**Ref.**	**Competence Statements**

LOGIC		
Propositions and connectivity.	D2p1	Know and understand how to form compound propositions by using ~, $\wedge$, $\vee$, $\Leftrightarrow$ and $\Rightarrow$.
	2	Be able to use truth tables to analyse propositions.
Switching and combinatorial circuits.	3	Be able to model compound propositions with simple switching and combinatorial circuits.
Boolean algebra.	4	Be able to manipulate Boolean expressions involving $\sim$, $\wedge$ and $\vee$ using the distributive laws, de Morgan's law, etc.

DECISION TREES		
Using networks in decision analysis.	D2N1	Be able to construct and interpret simple decision trees.
	2	Be able to use expected monetary values (EMVs) to compare alternatives.
	3	Be able to use utility to compare alternatives.

DECISION MATHEMATICS 2, D2			
Ref.	**Notes**	**Notation**	**Exclusions**

LOGIC			
D2p1		Propositions will be represented by lower case alphabetic characters.	
2			More than 3 propositions.
3	Logic gates: 		
4	Candidates will be expected to use (Boolean) algebraic manipulation to prove that simple Boolean expressions are equivalent.		

DECISION TREES			
D2N1			
2	Candidates will need to be able to distinguish between, and handle, decision and chance nodes.	EMV.	Explicit knowledge of Bayes’ theorem.
3		Decision nodes: Chance nodes:	